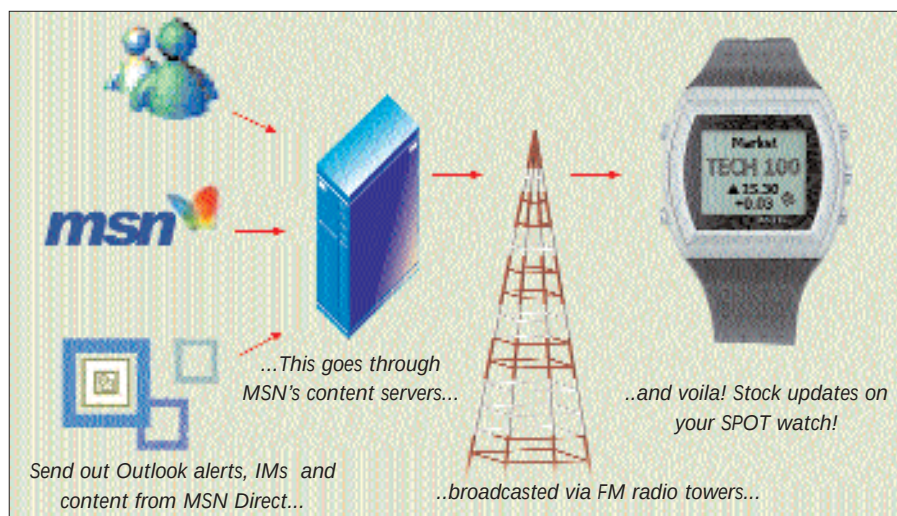




James Bond knows that, so does Mr India. Interested customers—called ‘early adopters’ in industry parlance—are signing up for the service in droves. Alan Reiter, wireless industry analyst and famed tech-blogger (<http://reiter.weblogger.com>—rated by Forbes magazine as one of the top five of its kind), is one such early adopter. Customers buy the watches from Fossil and Suunto—they retail for between \$100 and \$300—and then subscribe to MSN Direct (<http://direct.msn.com>) for \$9.95 a month or \$59 a year.

Indian customers will have to wait a while—Microsoft says they have no immediate plans of introducing the service in India.

When the service is off, or you go out of range, it's a normal watch; the rest of the time you get personalised content feeds straight from MSN Direct on the watch face. The watch receives data using an ingenious channel of data propagation, FM radio. This communication is strictly one-way; the watch receives data from MSN's content servers, but doesn't talk back to them.

Microsoft says that they researched the reactions of over 12,000 customers and found that most people prefer quick, glanceable information against two-way, immersive communication.

Binary tick tock

So what does a wrist-top fantasy look like? It has a slightly large build; it doesn't compete with the slim and elegant offerings from Swiss timekeepers, but it's not much different from the average sports watch. Reiter finds the watches to be slightly large

and clunky. He admits it's fun, but maintains that it's still for geeks and early adopters. Reiter should know, having used the Suunto N3 SPOT watch for quite a while now.

On the inside, the smart watch is actually a small wrist-top computer. The computing platform works on a seven-chip custom chipset, which measures 34 x 30 x 2 mm. Its main components are a baseband processor built on the core of an industry standard Arm7 processor that is commonly used in low-power applications.

According to National's Centrum Buero Information Telekommunikation (CeBIT) 2003 press release, the custom-designed integrated circuit (IC) runs at 28

MHz and processes the received signals. The baseband processors come built with digital signal processor (DSP) accelerators and have a unique 256-bit identification number etched onto the Arm7 core.

The other ICs on this system board include the sensitive Radio Frequency (RF) chip, which receives the FM radio signals. The RF chip can receive FM signals across the entire range of the commercial FM radio spectrum, from 88 to 110 MHz. The entire solution lives off 512 KB of ROM and 384 KB of RAM, which provides for several days of alerts and alarms apart from continuous updates of all

Info

The Arm7 chips used in SPOT watches are designed at National's India Design Center, located at Bangalore.

HOLLYWOOD

Watches in most Hollywood movies are just supremely expensive jewellery—from Bvlgari, Cartier, Rolex, to Omega.

However, a handful of movies have shown watches to be something more than just classy time keepers.

James Bond movies have shown the debonair spy wearing multi-functional devices in his Rolex and Omega.

In 1983's *Octopussy*, 007 sports a Seiko Sports watch with a TV screen; in *Golden Eye*, *Tomorrow Never Dies* and *The World Is Not Enough*, Bond sports classy Omegas that have remote detonators, laser metal cutters, and grappling hooks.

American cartoon fantasy character, Dick Tracy, had a two-way radio built into his watch. Not to be left out, Indian cinema had Mr India with a fantasy watch that could make the person wearing it invisible!

Out in movieland, a watch is certainly not just for telling time!



sorts. Power requirements for the system are proprietary information, but we found that SPOT watches use a unique induction charging mechanism that needs the watch to be placed on top of the charger; a full charge should suffice for three to five days.

The SPOT system's transmission protocols and architecture are just as inter-

